



Real World Multimodal Architectures



Code

- <https://colab.research.google.com/drive/1U5F7Qxo1gsLRk28fzH5Ek7qtLMXwRGS?usp=sharing>
- <https://colab.research.google.com/drive/1mQOi08E2Ldprm4jr6kAl4w4axZbOojLy?usp=sharing>
- <https://colab.research.google.com/drive/1IPykvlrOzzbaORbqwk5DgGGmtAVfCYUA?usp=sharing>

What will I learn?

1. CLIP
2. BLIP
3. Comparison of CLIP and BLIP
4. Diffusion

WHY?

- **WHY do we want to learn about architectures when all we need to build real world products is APIs (openai, anthropic, grok whatever)?**
- For some folks this can be totally unnecessary - just call apis!
- But for an “applied ML engineer”, even if you do not know all the deets about all the archs, you should have a general level understanding of various modules, paradigms, training and architectural techniques etc.
- For eg. if you ask me to explain [ResNet](#) right now, i won't be able to do that (not a vision person). But give me 20 mins, and I will be able to explain all the math+all the intuition everything. 100% confident about that! 🖐️ We want to prepare for you this as well.



CLIP

Contrastive Language-Image Pretraining

Introduction to CLIP

- ◆ **CLIP:** It stands for Contrastive Language-Image Pretraining. It was developed by OpenAI to bridge the gap between text and images.
- ◆ **Core Idea:** It aligns image and text embeddings in a shared space using paired data. It enables zero-shot learning: performs tasks without task-specific training.
- ◆ **Key Features:** Learns from large-scale internet datasets with diverse image-text pairs. Supports a wide range of tasks, including retrieval, classification, and captioning.
- ◆ **Significance:** Foundation for many multimodal AI systems, driving advances in tasks like image generation and visual search.

Architecture

- ◆ **Image Encoder:** Uses models like ResNet or Vision Transformer to process images and extract feature vectors.
- ◆ **Text Encoder:** Typically a Transformer-based model (like GPT or BERT) processes text and generates embeddings.
- ◆ **Shared Embedding Space:** Both image and text embeddings are mapped into the same space to measure similarity.
- ◆ **Projection Heads:** Linear layers ensure that embeddings from both encoders are comparable in the shared space.

[\[Read More\]](#)

Working

◆ **Input:** Paired image-text data (e.g., a photo of a dog and the caption “A dog running on the beach”).

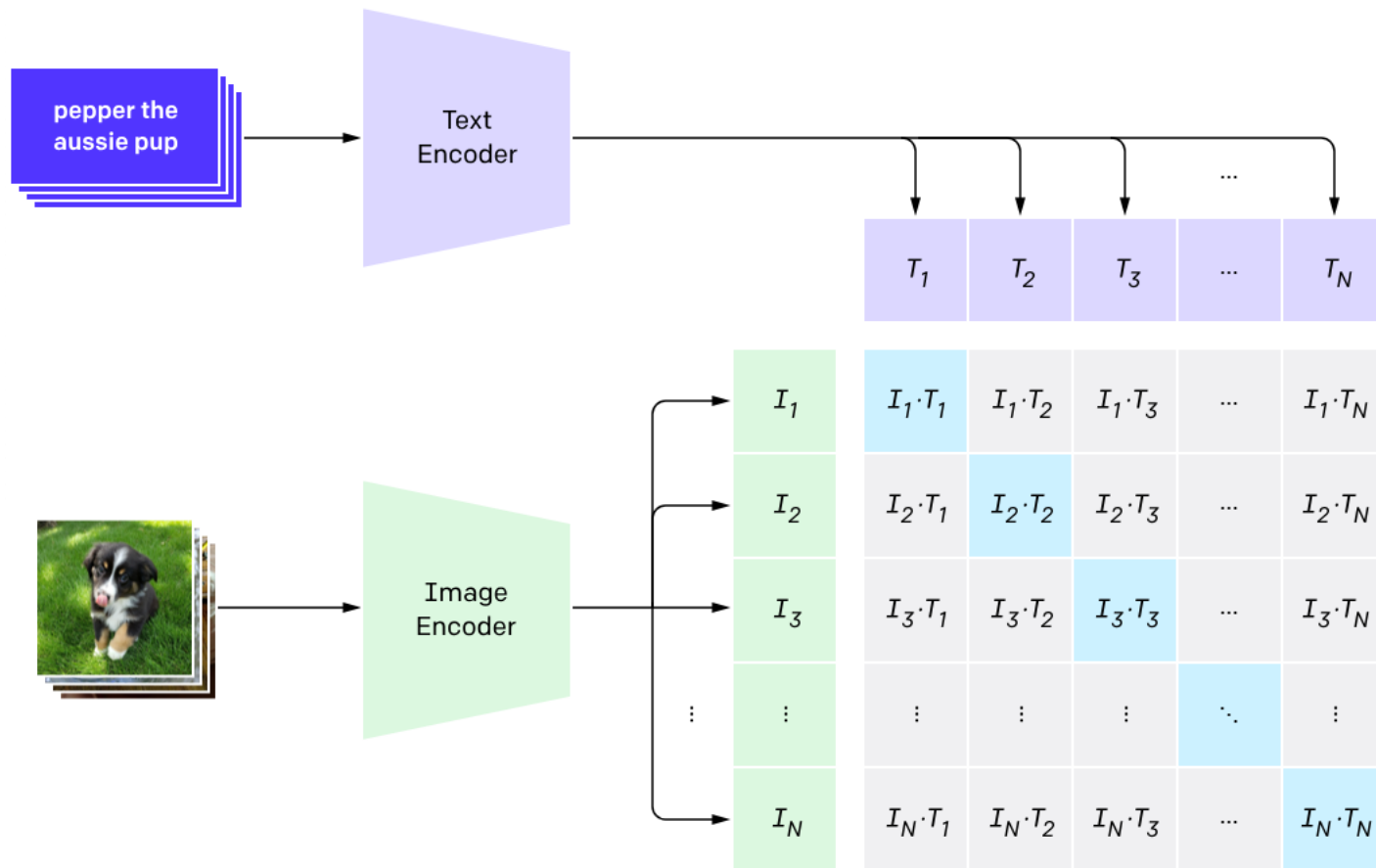
◆ **Encoding:** The image encoder extracts visual features, while the text encoder converts the caption into text features.

Objective: Given an image, predict which out of a set of 32,768 randomly sampled text snippets, was actually paired with it in our dataset.

◆ **Contrastive Loss:** Ensures that embeddings of matching pairs (image-text) are closer, while non-matching pairs are farther apart in the shared space.

Zero-shot Capability: During inference, the model can classify images or retrieve them based on unseen textual descriptions.

1. Contrastive pre-training

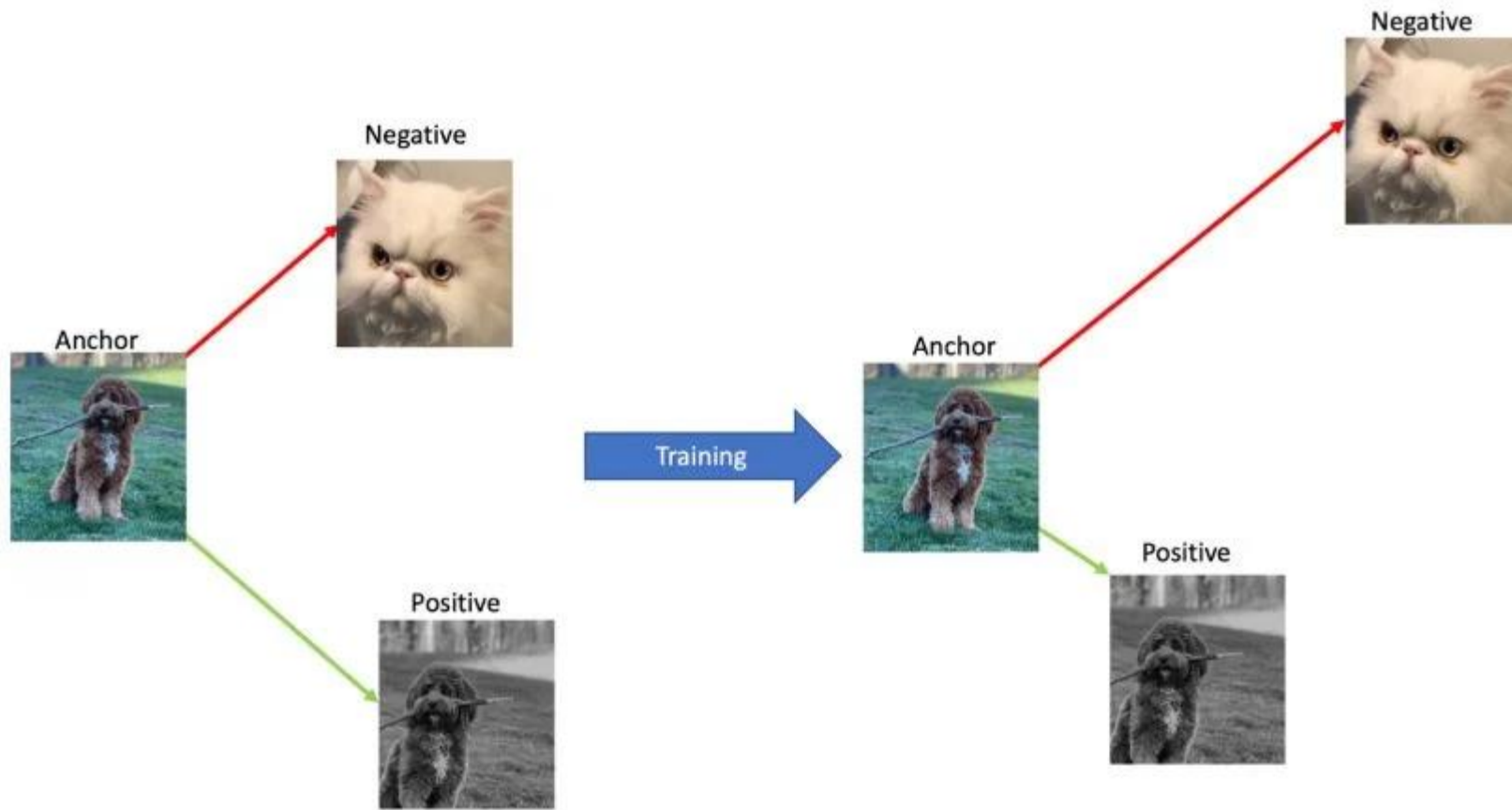


Contrastive Learning

◆ A learning technique where the model is trained to **bring similar pairs closer** in the embedding space and **push dissimilar pairs farther** apart. In CLIP, this is applied to **align** image and text embeddings effectively.

◆ **How it Works in CLIP:**

1. **Paired Data:** The model is trained on paired image-text data.
2. **Embedding Generation:**
 - Image encoder processes images to generate embeddings.
 - Text encoder processes captions to create text embeddings.
3. **Positive Pair:** Each image-text pair is treated as a positive example.
4. **Negative Samples:** All other image-text combinations in the batch act as negatives.
5. **Contrastive Loss:** Maximizes similarity for positive pairs and minimizes it for negatives.



Advantages/Disadvantages



Advantages:

- Generalizes well to many tasks without fine-tuning.
- Learns meaningful cross-modal representations.
- Scalable with large datasets.



Disadvantages:

- Requires extensive compute and large-scale data for training.
- Struggles with fine-grained understanding and contextual nuances.

Real World Applications

- ◆ [Manufacturing Quality Inspection \(Few-Shot\)](#)
- ◆ [Search for Design Patents](#)
- ◆ [Robotic CLIP: Action Recognition in Robotics](#)



BLIP

Bootstrapping Language-Image Pretraining

Introduction to BLIP

- ◆ **BLIP:** Stands for Bootstrapping Language-Image Pretraining. Aims to improve multimodal AI by combining alignment and generation tasks.
- ◆ **Core Idea:** Integrates supervised and self-supervised learning to utilize both labeled and unlabeled data. Refines multimodal representations through iterative bootstrapping.
- ◆ **Key Features:** Flexible architecture supporting contrastive learning, generation, and hybrid tasks. Adapts to diverse multimodal tasks, from image captioning to visual question answering.
- ◆ **Significance:** Advances multimodal understanding and generation, improving AI's ability to process and combine textual and visual data.

Architecture

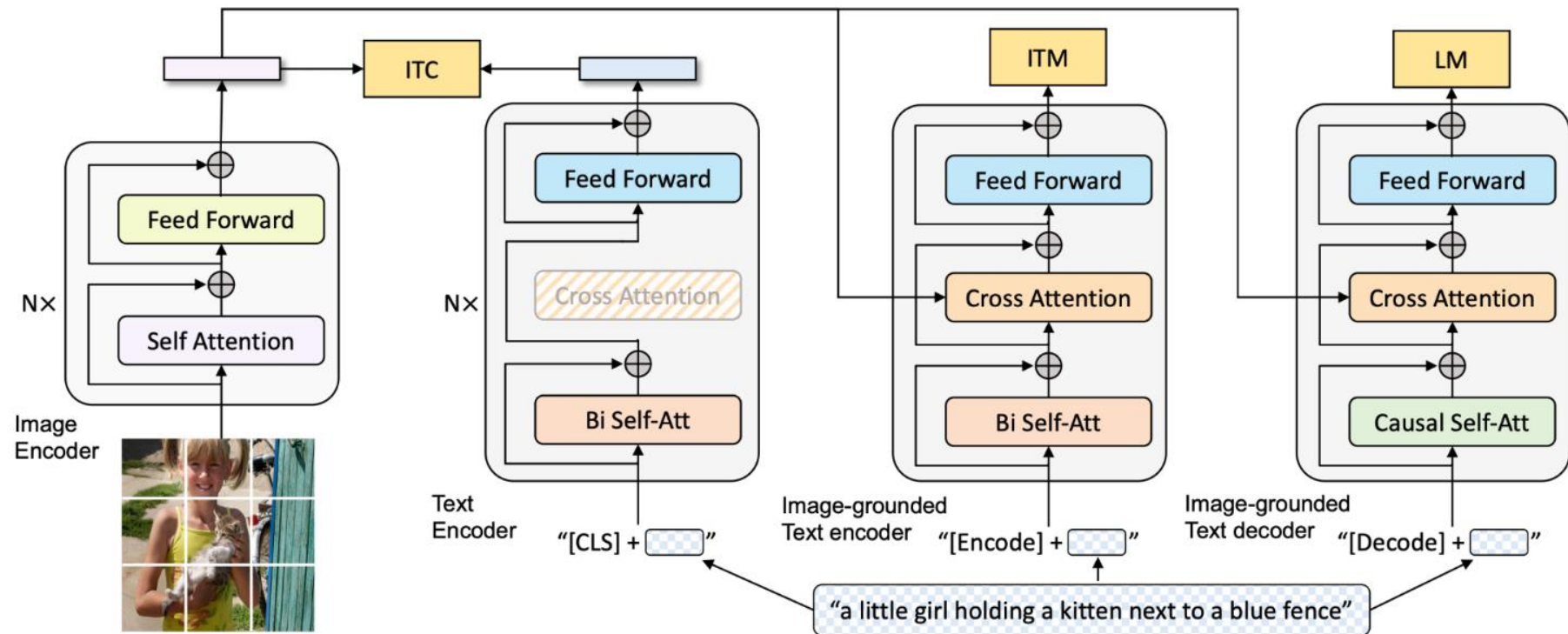
- ◆ **Vision and Text Encoder:** A Vision Transformer extracts features from input images. The Text Encoder does the same for corresponding texts.
- ◆ **Text Decoder:** A Transformer-based decoder generates or processes text sequences using images only; enabling tasks like captioning.
- ◆ **Multimodal Encoder:** Combines the outputs from the vision and text encoders to process and align both modalities simultaneously.
- ◆ **Bootstrapping Module:** Uses pseudo-labeling to iteratively improve performance on unlabeled data by refining predictions.

[\[Read More\]](#)

[\[Fine Tuning Code\]](#)

BLIP ARCHITECTURE

- ITC - image text contrastive loss
- ITM - image text matching
- LM - language modelling



Generate text = decoder

Encoder = more about forming a repr of the input

Decoder = decoding the info = generate the info

Working

- ◆ **Input:** Takes image-text pairs and unlabeled images to leverage both supervised and unsupervised learning.
- ◆ **Multimodal Training:**
 - **Contrastive Objective:** Aligns image-text pairs, similar to CLIP.
 - **Generative Objective:** Trains the text decoder to generate meaningful captions or responses based on images.
 - **Matching Objective:** Given a pair of image-text, predict if they match or not.
- ◆ **Iterative Refinement:** Unlabeled data is pseudo-labeled using the current model, and the predictions are refined with further training.
- ◆ **Flexibility:** Supports diverse tasks like retrieval (contrastive) and generation (decoder).

Advantages/Disadvantages

◆ Advantages:

- Flexible for multiple tasks (e.g., classification, captioning).
- Better at understanding complex relationships between modalities.
- Incorporates both alignment and generation, improving versatility.

◆ Disadvantages:

- Computationally intensive for training and fine-tuning.
- Less efficient for tasks requiring strict alignment, like retrieval.

Real World Applications

- ◆ [MedBLIP: Medical Image Captioning](#)
- ◆ [BLIPCam: Live Image Captioning](#)
- ◆ [Video Moment Retrieval](#)

Comparison of CLIP and BLIP

Aspect	CLIP	BLIP
Objective	Trained with <i>only</i> image–text contrastive learning to align modality embeddings.	Uses <i>multi-task objectives</i> : contrastive (ITC), matching (ITM), and generative captioning .
Architecture	Two separate encoders (image + text), mapped into a shared embedding space—no cross-attention during encoding .	Multimodal Mixture of Encoder-Decoder (MED) : includes unimodal encoders, cross-attentive encoder, and causal decoder .
Ability to Generate Text	✗ No built-in text generation—works for retrieval and classification.	✓ Supports image captioning and VQA via text decoder.
Handling Noisy Data	Relies on large-scale web pairing without explicit noise filtering.	Uses synthetic captions and <i>CapFilt</i> filtering to clean noisy web data .
Use Cases	Best for zero-shot classification and text-image retrieval.	Better suited for retrieval <i>and</i> generation (captions, VQA), with more fine-grained understanding.

Fusion Strategy of CLIP and BLIP

Aspect	CLIP (Late Fusion)	BLIP (Hybrid Fusion)
Fusion strategy	Processes image and text separately, then aligns final embeddings via dot product.	Uses cross-attention in encoder for joint processing + late fusion via the decoder.
Architectural setup	Independent image encoder and text encoder; no interaction until embedding comparison.	Multimodal encoder-decoder (MED) with cross-attentive layers in mid-stream .
Cross-modal flow	Fusion only at end—final dot-product similarity for retrieval/classification.	Fusion occurs during encoding (intermediate) and again during decoding (late).
Strengths	Simple, fast, modular; easy to scale and adapt.	Rich interaction across modalities; supports both alignment and text generation.
Trade-offs	Limited to retrieval tasks; lacks detailed cross-modal context.	More complex and computation-heavy due to cross-attentive modules.



Diffusion

Diffusion is the movement of particles from a region of higher concentration to a region of lower concentration.

FORWARD PROCESS



REVERSE PROCESS



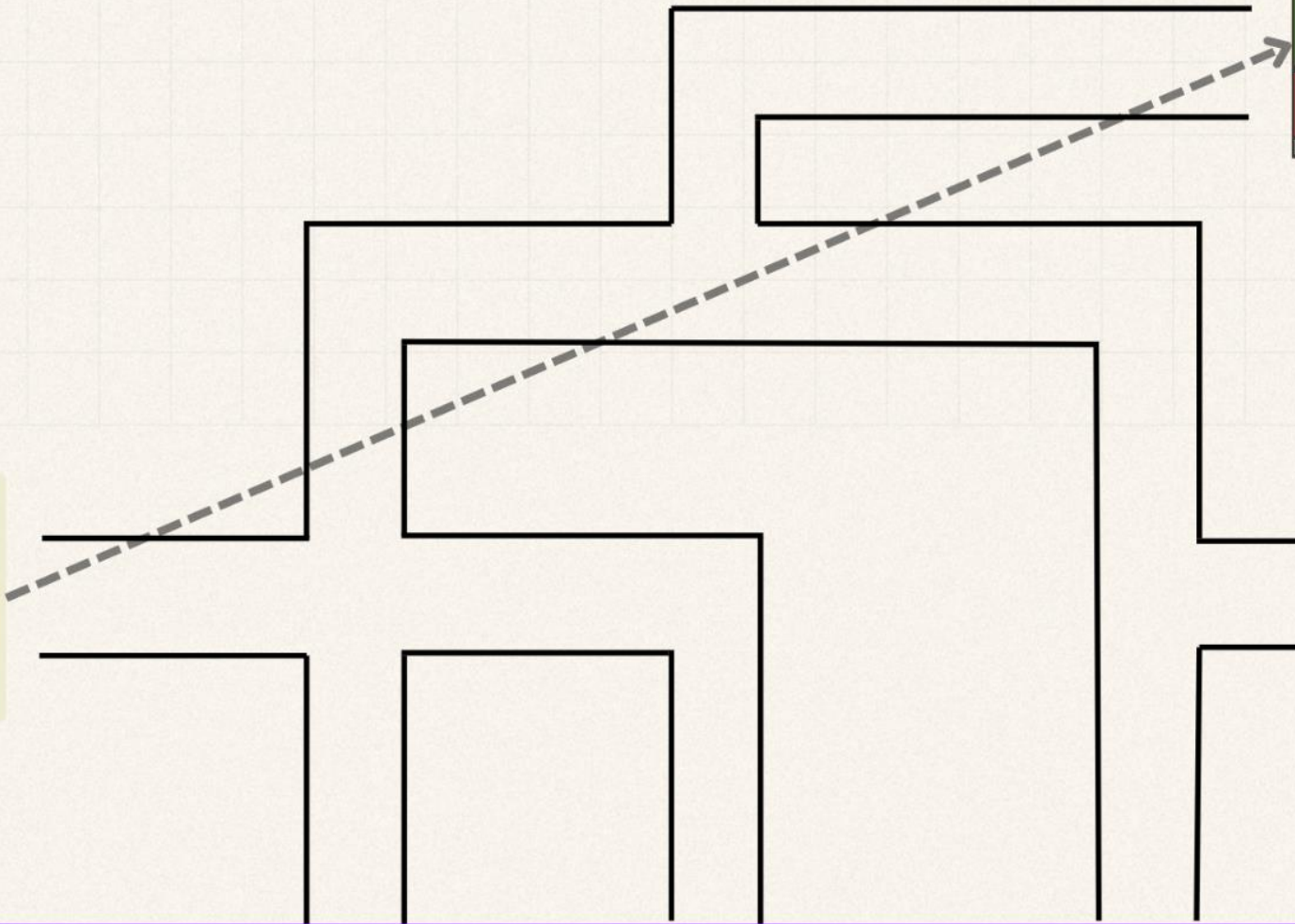
Forward Process

FORWARD PROCESS



FORWARD PROCESS







GO STRAIGHT →

↑ TAKE A LEFT ↓

→ GO STRAIGHT →

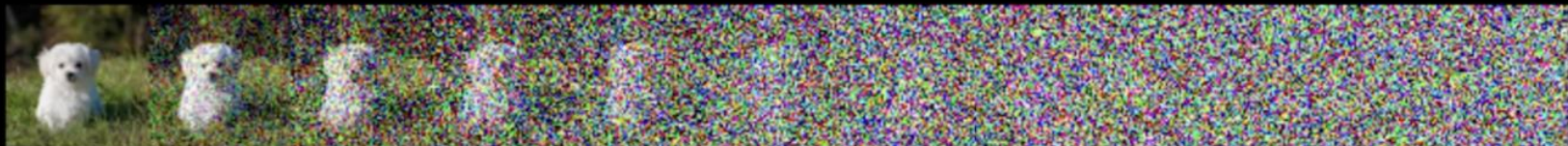
↑ LEFT ↓

→ GO STRAIGHT →



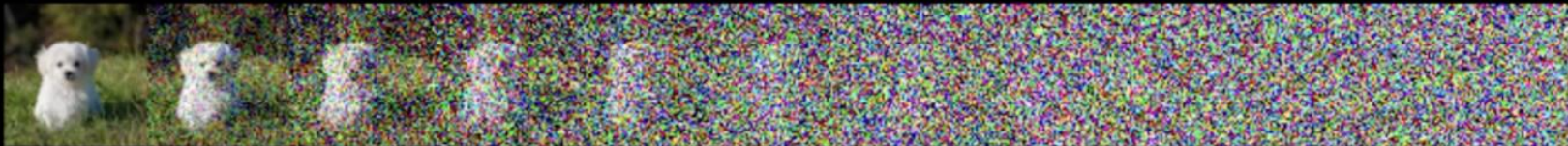
Noise Schedule

linear



Noise Schedule

linear



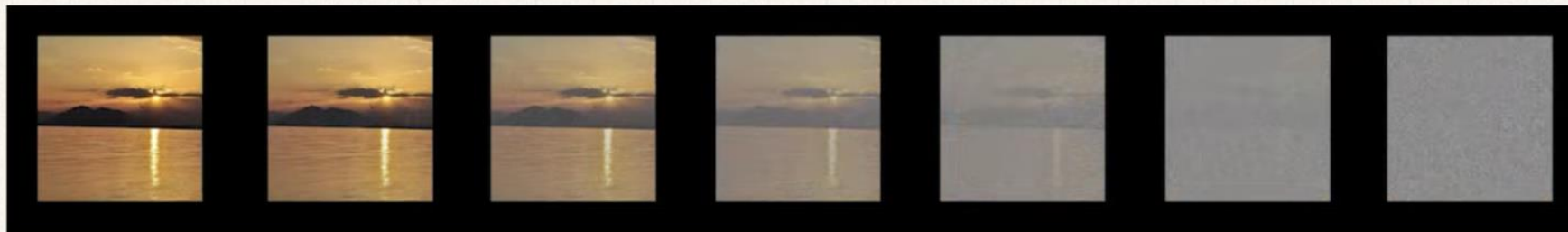
cosine



Reverse Process

PROMPT

← REVERSE PROCESS →





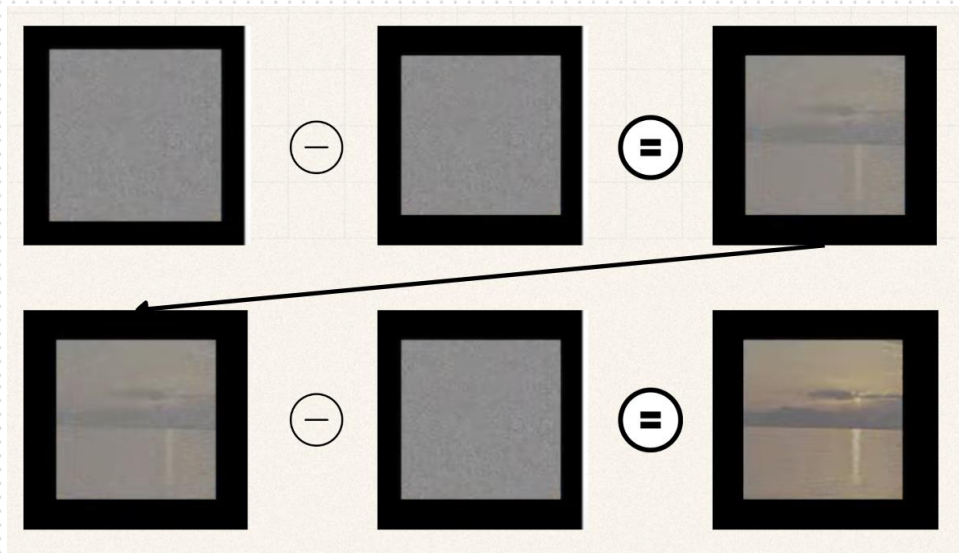
Input noise



Noise to be
removed



Image after noise
removal

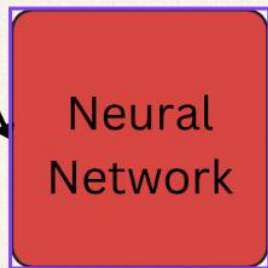




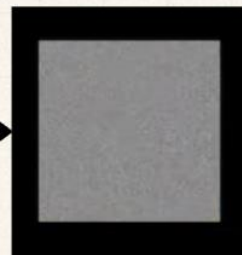




Input Image



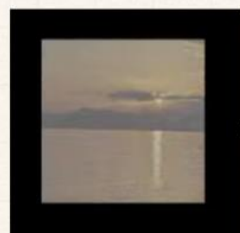
Neural
Network



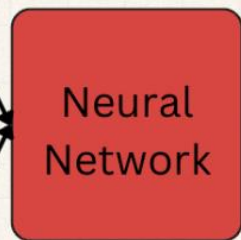
Noise to be
Subtracted



Clearer Image

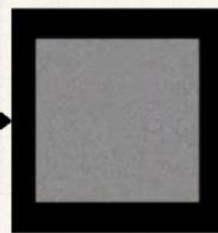


Input Image



Neural
Network

Conditioning



Noise to be
Subtracted



Clearer Image



Embedding

Prompt: A sunset over a calm lake with mountains in the background and the sun reflecting on the water.

Diffusion in Real Life

“

Text to Image

- DALL-E
- Stable Diffusion



“

Text to Video

- Luma AI
- Runway ML



“

Image to Image

- Stable Diffusion (Inpainting)
- ControlNet

ControlNet For Stable Diffusion AI



“

Text to Audio

- Suno AI
- AudioGen
- Dance Diffusion

SUNO

AI Music Creator

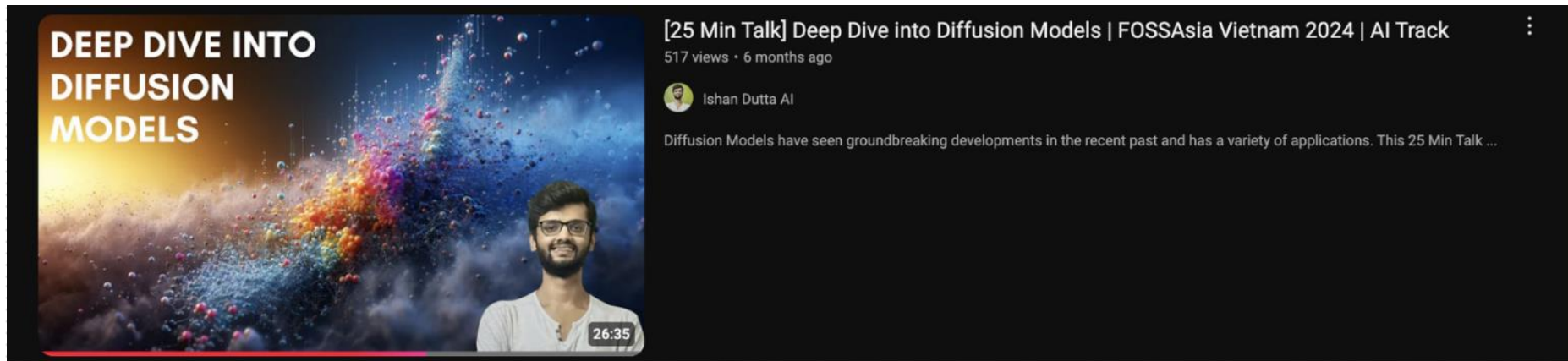
“

Text to 3D

- Dream Fusion
- Point-E



Maths behind Diffusion Models



[\[YouTube Link\]](#)

Lilian weng's blog -

<https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>